

Transfer functions

R0005E - Measurement and Feedback Control

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Representations of a transfer function

When the transfer function is derived from an input-output form of the differential equation we usually get the following result:

$$H(s) = \frac{b_m s^m + \dots + b_1 s + b_0}{a_n s^n + \dots + a_1 s + a_0}$$

with $m \leq n$.

Note: It is assumed that all initial conditions are zero!

We know from algebra that the roots of the numerator and denominator polynomial are either real numbers or complex conjugate pairs of complex numbers.

Therefore the transfer function can be represented in the so called zero-pole-gain representation given in the form:

$$H(s) = K \frac{\prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)}$$

where K is the gain, z_j are the zeros and p_i are the poles.

Applying the partial fraction expansion on the zero-pole-gain representation and assuming only distinct poles yields

$$F(s) = \sum_{i=1}^n \frac{A_i}{s - p_i}$$

Thus, the time function $f(t)$ is the superposition of time functions relating to the individual poles. These individual addends are the **modes** of the transfer function and the system.

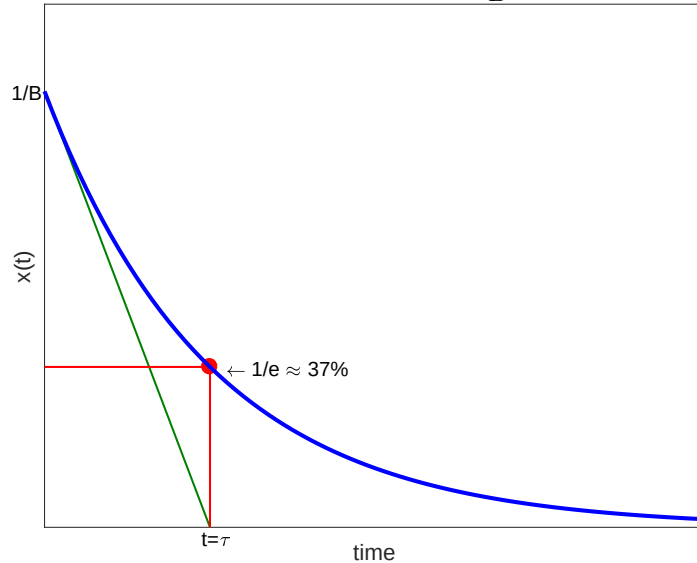
Effect of pole locations

We return to the spring-damper example. The transfer function was determined to be

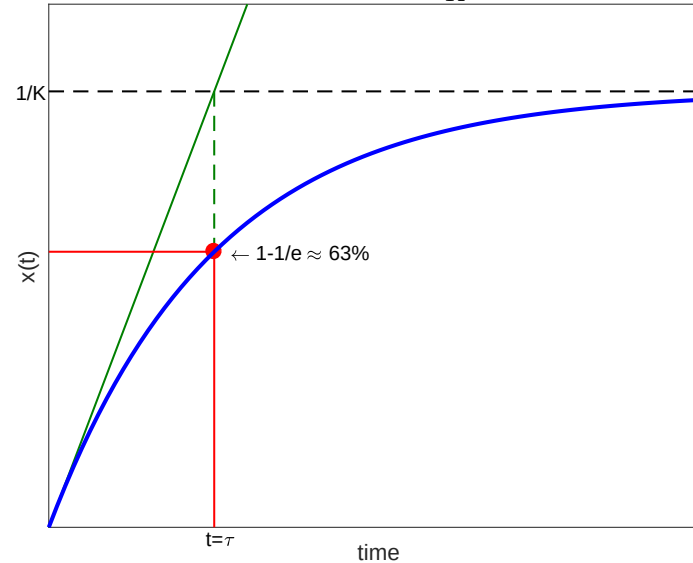
$$H(s) = \frac{1/B}{s + K/B}$$

The response $x(t)$ of this first-order system for an impulse and a step are

$$u(t) = \delta(t), \quad x(t) = \frac{1}{B}e^{-\frac{K}{B}t}$$



$$u(t) = \mathbf{1}(t), \quad x(t) = \frac{1}{K}(1 - e^{-\frac{K}{B}t})$$

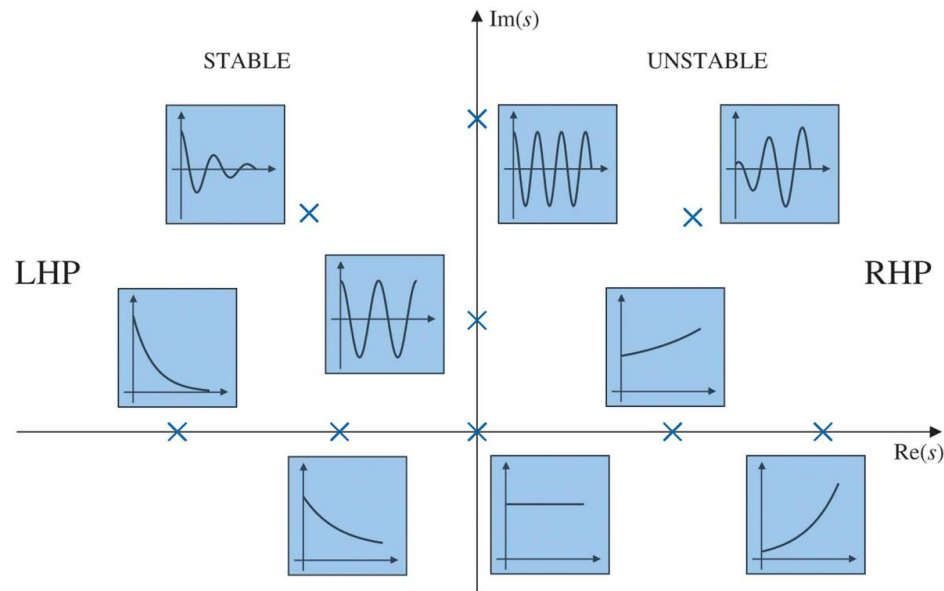


These plots are derived for $K = 1\text{N}/m$ and $B = 1\text{Ns}/m$. It can be shown that the time constant $\tau = \frac{B}{K}$. At the same time $p = -\frac{K}{B}$ is the pole location of the pole of the transfer function.

Thus, $\tau = -\frac{1}{p}$.

Clearly, the pole location in the complex s -plane determines the type of response and how fast the transients are decaying:

- Far left: fast decay.
- Non-zero imaginary part: oscillations.
- Right half plane: unstable.



Source: Feedback Control of Dynamic Systems, Franklin et. al.

In contrast to a first order system, second order systems can exhibit oscillations. These oscillations are due to the imaginary parts of the poles.

Let's start from the linear differential equation

$$a_2\ddot{y} + a_1\dot{y} + a_0y = b_1\dot{u} + b_0u$$

Applying the Laplace transform and considering the initial conditions yields

$$a_2(s^2Y(s) - sy(0) - \dot{y}(0)) + a_1(sY(s) - y(0)) + a_0Y(s) = b_1(sU(s) - u(0)) + b_0U(s)$$

We collect the terms and resolve for $Y(s)$ and get

$$Y(s) = \frac{b_1s + b_0}{a_2s^2 + a_1s + a_0}U(s) + \frac{(a_2s + a_1)y(0) + a_2\dot{y}(0) - b_1u(0)}{a_2s^2 + a_1s + a_0}$$

Note:

- In both parts, we have the same denominator. The denominator is also called the **characteristic polynomial**.
- From the characteristic polynomial the poles of the differential equation and the system can be derived by solving the algebraic equation for s

The poles of the above second order system are given by:

$$p_{1,2} = -\frac{a_1}{2a_2} \pm \sqrt{\frac{a_1^2}{4a_2^2} - \frac{a_0}{a_2}}$$

There are now two cases: (1) Two real poles, (2) a pair of conjugate complex poles.

Case 1: Two real poles p_1 and p_2

The zero state response with $u(t) = \delta(t)$ can now be rewritten using the partial fraction expansion:

$$\begin{aligned} Y_{zi}(s) &= \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + a_0} \\ &= \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} \\ &\Downarrow \\ y_{zi}(t) &= K_1 e^{p_1 t} + K_2 e^{p_2 t} \end{aligned}$$

The response is stable if $p_1 < 0$ and $p_2 < 0$.

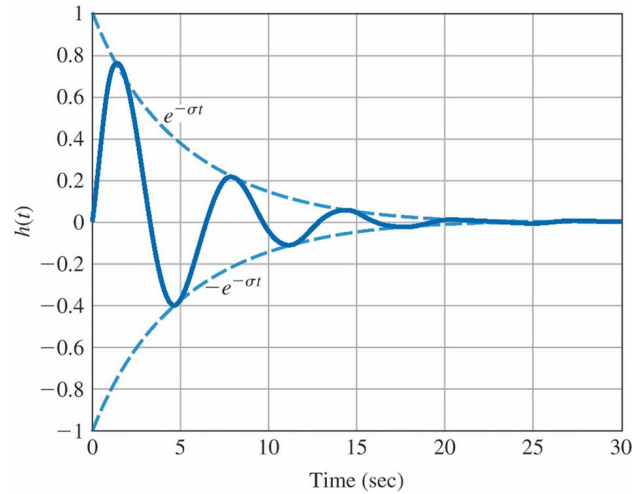
Case 2: p_1 and p_2 are a complex conjugate pair

Then, the poles can be written as $p_{1,2} = -\sigma \pm j\omega_d$ and the partial fraction expansion of the zero state response with $u(t) = \delta(t)$ results in

$$\begin{aligned} Y_{zi}(s) &= \frac{K e^{j\phi}}{s + \sigma - j\omega_d} + \frac{K e^{j\phi}}{s + \sigma + j\omega_d} \\ &\Downarrow \\ y_{zi}(t) &= 2K e^{-\sigma t} \cos(\omega_d t + \phi) \end{aligned}$$

Obviously, the term $e^{-\sigma t}$ determines the stability of the response. ϕ considers the initial conditions.

The response for $\phi = -\frac{\pi}{2}$ could be the following:



Source: Feedback Control of Dynamic Systems, Franklin et. al.

Some conclusions:

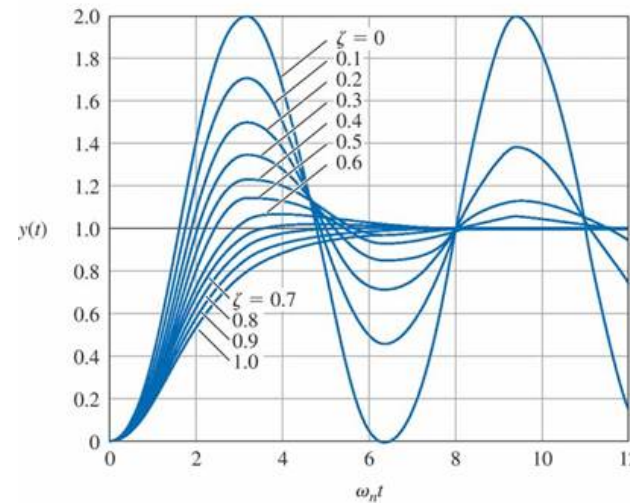
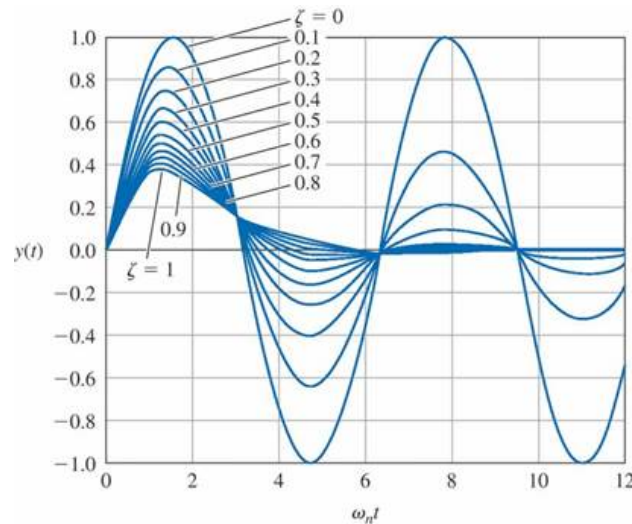
- Stability is determined by σ . It has to be positive.
- Stability requires that the poles of a transfer function have negative real parts.

In the response figure of the second order system, we could see that the oscillation decays. This decay can be characterized by a **damping coefficient (damping ratio)** ζ . Additionally, one could think that the system would oscillate freely when the damping coefficient is 0. This frequency is called the **(undamped) natural frequency** ω_n .

If a second order transfer function has complex poles, then the characteristic polynomial can be written in the form:

$$s^2 + 2\zeta\omega_n s + \omega_n^2,$$

where $\sigma = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$. The system will oscillate at the damped frequency ω_d .

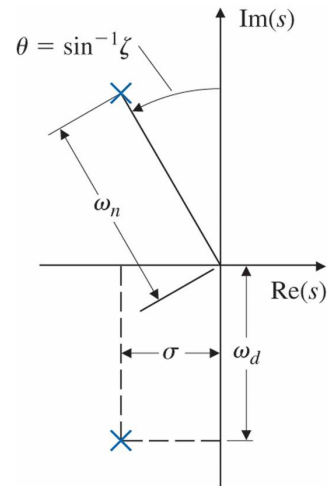


Source: Feedback Control of Dynamic Systems, Franklin et. al.

Similarly, the complex conjugate pole pair can be given in terms of the damping coefficient and the natural frequency:

$$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

This leads to the following location plot in the complex plane:



Note: For $\zeta \geq 1$ the poles become real.

Terminology:

1. Undamped: $\zeta = 0$
2. critically damped: $\zeta = 1$

Special case: $p_1 = p_2$

From the partial fraction expansion we know that multiple poles at the same location are treated differently than distinct poles.

The Laplace transform for multiple poles is as follows:

$$f(t) = t^n e^{at} \quad \Rightarrow \quad \mathcal{L}(f(t)) = \frac{n!}{(s - a)^{n+1}}$$

The zero state response with $u(t) = \delta(t)$ for the special case of $p_1 = p_2$ is then

$$\begin{aligned} Y_{zi}(s) &= \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + a_0} \\ &= \frac{K_1}{s - p_1} + \frac{K_2}{(s - p_1)^2} \\ &\Downarrow \\ y_{zi}(t) &= K_1 e^{p_1 t} + K_2 t e^{p_1 t} \\ &= (K_1 + K_2 t) e^{p_1 t} \end{aligned}$$

Again, the response is stable for $p_1 = p_2 < 0$, but the factor $(K_1 + K_2 t)$ will delay the decay of the response to zero.

DC-Gain

The poles affect the shape of the response during the transient behavior at large. Additionally, to the transient behavior there is the steady state or static behavior.

We know that the output of an arbitrary system is given by

$$Y(s) = H(s)U(s) \quad (1)$$

Assume that

- $H(s)$ is stable
- $U(s)$ is the Laplace transform of a unit step

Then we can calculate the steady state output by applying the final value theorem to (1) and get

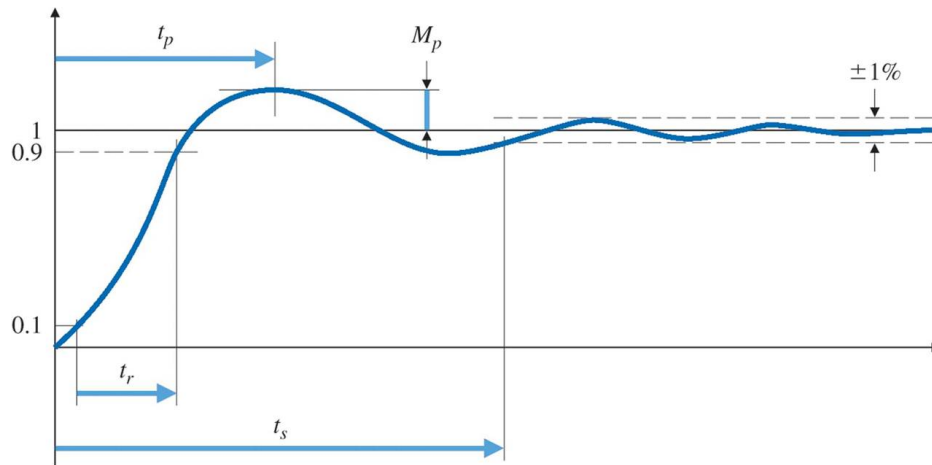
$$\begin{aligned} y_{ss} &= \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sH(s) \frac{1}{s} \\ &= H(0) \\ &= y(t)|_{t \rightarrow \infty} \end{aligned}$$

Thus, $H(0)$ is the gain factor for the amplitude of the input signal for $t \rightarrow \infty$. It is called the **DC-Gain**.

Time domain specifications

We have seen that the poles of a system dictate the transient behavior of the system. But it is hard to understand directly how the location of a pole can be interpreted in terms of the behavior in the time domain.

Let's assume the dynamic response of a system has the following shape



Questions:

- Imagine your requirements on a cruise control?
- Would this be acceptable?
- How could the behavior look like?

Most importantly, there are certain parameters in the plot, that can be used to characterize the requirements:

- **Rise time** t_r : The time it takes for the system to rise from 10% of the desired output to 90% of the desired output.
- **Settling time** t_s : The time it takes for the transients of the system to decay to a certain degree.
- **Overshoot** M_P : The maximum amount the system exceeds the final value relative to the final value. Often given in percent.
- **Peak time** t_p : The time at which the maximum overshoot is reached.

Idea:

Relate these parameters with coefficients of a transfer function! The gain, the location of the poles, the natural frequency and/or the damping coefficient.

Result:

We could design a system (controller and process) such that its transfer function complies with the requirements!

Rise time:

For the rise time, no generally valid relationship giving the transfer function parameters can be given. There is only a coarse approximation that can be used for second order systems with a damping of $\zeta = 0.5$:

$$t_r \cong \frac{1.8}{\omega_n}$$

Still, it is known that a larger ω_n will yield a shorter rise time.

Overshoot and peak time

In contrast to the rise time, the overshoot M_P and the peak time t_p can be analytically derived. Clearly, M_P is an extremum of the response function and the peak time is the time of its occurrence.

Thus, the derivative of the response function can be assessed and results in the following formulas for a second order system:

$$t_p = \frac{\pi}{\omega_d}$$
$$M_P = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}, \quad 0 \leq \zeta < 1$$

Settling time

The settling time can be derived depending on the relative accuracy that need to be achieved. Typical values are 5%, 2% and 1%.

Remember that the response function of a second order system is given by

$$y(t) = 2Ke^{-\sigma t} \cos(\omega_d t + \phi)$$

with $\sigma = \zeta\omega_n$.

Clearly, the settling time is determined by the term $e^{-\zeta\omega_n t_s}$ and for the different levels of accuracy we get:

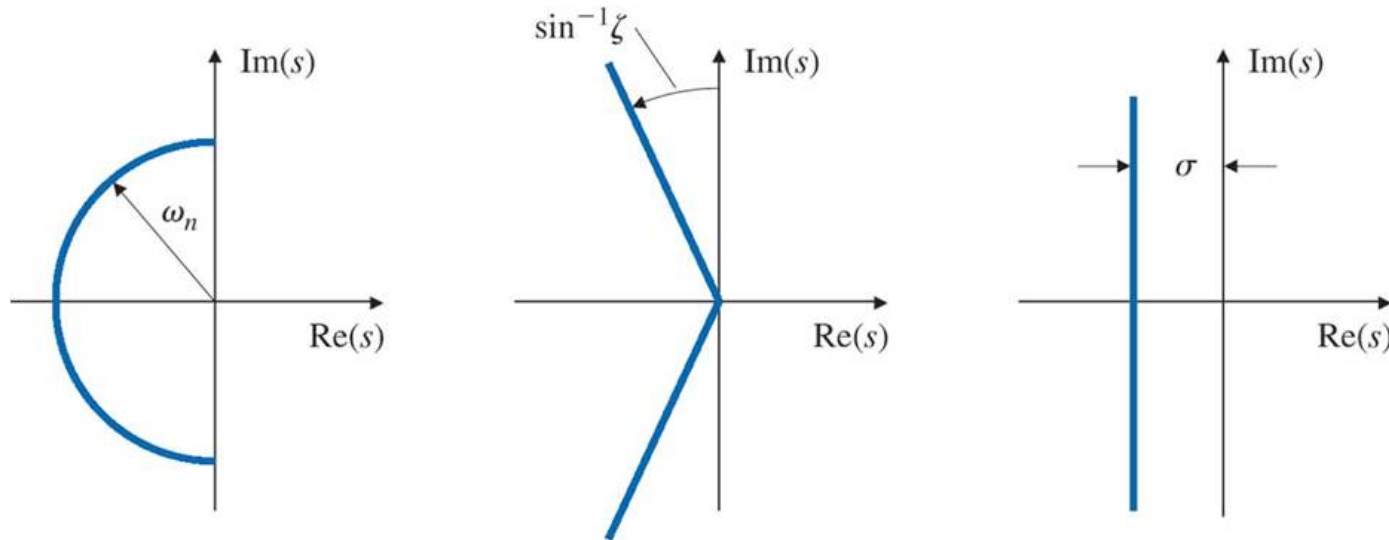
$$\begin{aligned} 5\% & \Rightarrow t_s = \frac{3}{\zeta\omega_n} \\ 2\% & \Rightarrow t_s = \frac{4}{\zeta\omega_n} \\ 1\% & \Rightarrow t_s = \frac{4.6}{\zeta\omega_n} \end{aligned}$$

Recommended problem:

- Problem 3.23 lets you practice the design of a component in a system
- Problem 3.24 lets you practice the design of a controller gain in a closed loop system.

Specification in the Laplace domain

Now that the relationship to the parameters ζ , ω_n and $\sigma = \zeta\omega_n$ are established it is possible to determine regions in the complex s-plane.



t_r implies minimum ω_n and the poles have to be placed outside the half circle.	M_P implies a minimum ζ and the poles have to be placed inside the cone.	t_s implies a minimum decay rate σ and the poles have to be placed left of $-\sigma$.
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Conclusion: Now we can relate time domain parameters with transfer function parameters without calculating the solutions to the differential equations.

This is the second step towards the design of a controller!

Recommended problem: 3.27 lets you practice the use of design regions.

Effect of an additional zero

Now that we understand the effect of poles on a system behavior, the natural question is: What effect do have zeros or additional poles?

Let's have a look at the following example:

$$H(s) = \frac{50(s + a)}{a(s + 10)(s + 5)}$$

This second order system has a DC-Gain of $H(0)=1$. Thus, at steady state it will always attain the input step amplitude.

The poles of this system are $p_1 = -5$ and $p_2 = -10$. Additionally, there is a zero at $z_1 = -a$.

We can now apply the partial fraction expansion on $H(s)$ and get

$$H(s) = \frac{50(s + a)}{a(s + 10)(s + 5)} = \frac{10 \frac{10-a}{a}}{s + 10} + \frac{10 \frac{a-5}{a}}{s + 5}$$

Note:

The superposition indicates and quantifies

- the affect of the zero.
- the relative contribution of each addend.

Now we can identify several cases:

- $a = 5$ or $a = 10$:

One of the addends diminishes and the transfer function can be reduced to either

$$H(s) = \frac{5}{s+5} \quad \text{or} \quad H(s) = \frac{10}{s+10}$$

This is also called a pole-zero cancellation. Already in the close proximity of the cancellation, the effect of that addend is becoming quite small.

- $a \approx 0$:

It means that the coefficients in the superposition become very large with opposite signs. Remember that the DC-Gain of the system is still 1. There should be a large overshoot!

- $a \rightarrow \infty$:

The affect of the zero diminishes and the system becomes a second order system without a zero.

$$\Rightarrow H(s) \approx \frac{50}{(s+10)(s+5)}$$

- $a < 0$:

Now we get a zero in the right half plane. In this case one of the addends gets a negative sign. Which means that the system should start in the wrong direction.

This type of zero is also called **non-minimum phase zero**.

We have a closer look at the behavior when a unit step input occurs and what the resulting steady state will be

Final value:

We can apply the final value theorem at this case and get

$$\begin{aligned}y(t \rightarrow \infty) &= \lim_{s \rightarrow 0} sH(s)U(s) \\&= s \frac{50(s+a)}{a(s+10)(s+5)} \frac{1}{s} \\&= 1\end{aligned}$$

Thus, the output follows the step at steady state no matter where the zero is placed.

Slope at the occurrence of the step:

The slope indicates in which direction the response starts and how "intense" the response will be.

$$\begin{aligned}\dot{y}(0) &= \frac{dy}{dt}(0) \\&= \lim_{s \rightarrow \infty} s^2 H(s)U(s) \\&= \lim_{s \rightarrow \infty} s^2 \frac{50(s+a)}{a(s+10)(s+5)} \frac{1}{s} \\&= \frac{50}{a}\end{aligned}$$

Stability

There are two types of stability definition:

- **Bounded-input bounded-output stability:**

A system with impulse response $h(t)$ is BIBO stable if and only if

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

- **Internal stability:**

A system with impulse response $h(t)$ is stable if all the poles of the transfer function $H(s)$ have negative real parts, and is unstable otherwise.

Note: For internal stability, make use of the partial fraction expansion and examine the stability of each addend. If there is an addend which is not stable, then the complete system is unstable.

There is a condition on the characteristic polynomial that gives quick information on stability.

Necessary condition for stability:

All the coefficients of the characteristic polynomial are positive.